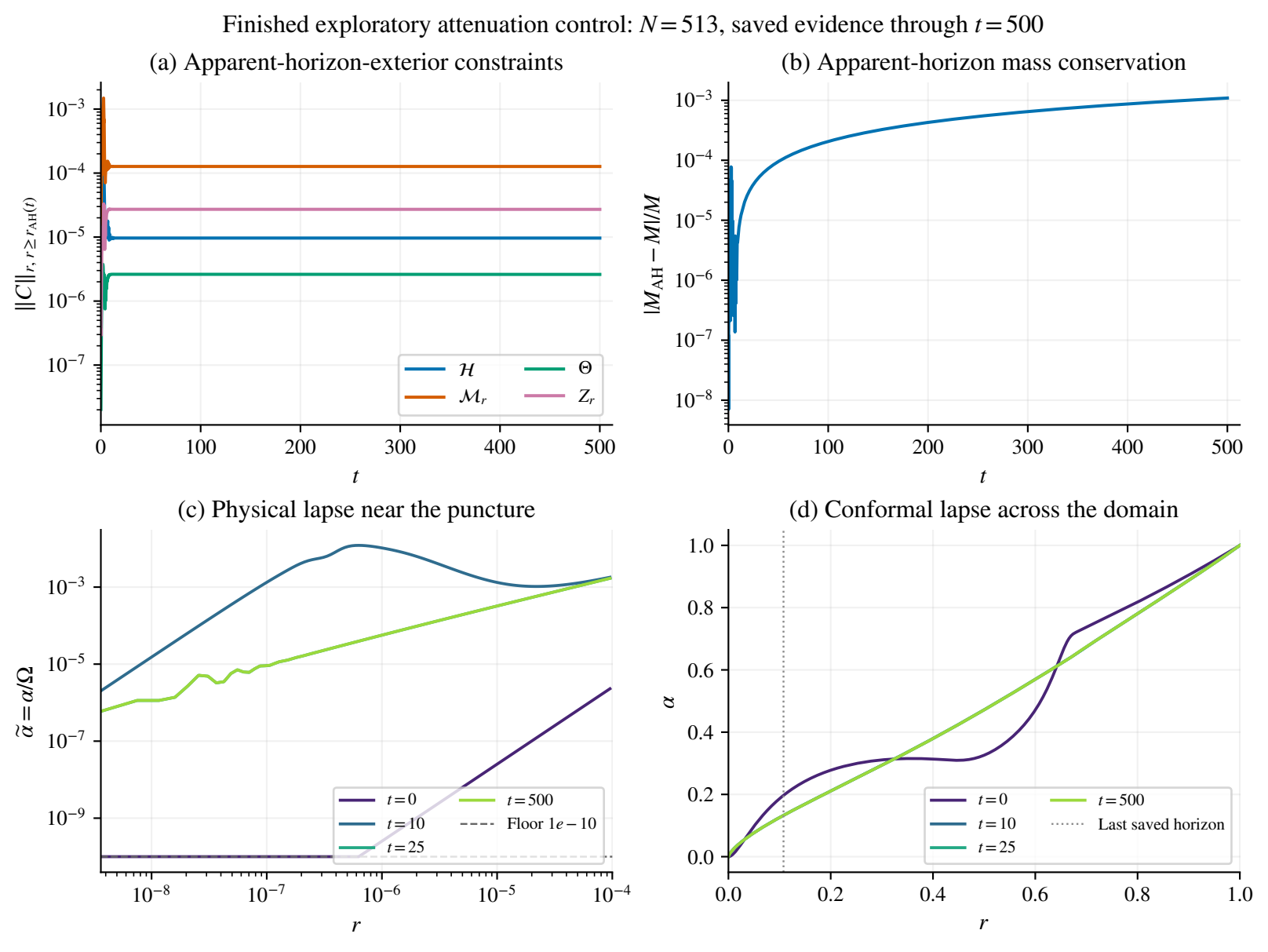
Finished exploratory attenuation control: $N = 513$, saved evidence through $t = 500$
(a) Apparent-horizon-exterior constraints
$\|C\|_{r,\, r \geq r_{\mathrm{AH}}(t)}$
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
$\mathcal{H}$
$\mathcal{M}_r$
Θ
Z_r
0
100
200
300
400
500
t
(b) Apparent-horizon mass conservation
$|M_{\mathrm{AH}} - M|/M$
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
0
100
200
300
400
500
t
(c) Physical lapse near the puncture
$\widetilde{\alpha} = \alpha/\Omega$
10^{-3}
10^{-5}
10^{-7}
10^{-9}
$t = 0$
$t = 10$
$t = 25$
$t = 500$
Floor $1e-10$
10^{-8}
10^{-7}
10^{-6}
10^{-5}
10^{-4}
r
(d) Conformal lapse across the domain
α
1.0
0.8
0.6
0.4
0.2
0.0
$t = 0$
$t = 10$
$t = 25$
$t = 500$
Last saved horizon
0.0
0.2
0.4
0.6
0.8
1.0
r